

Che 205 Cover Sheet

Name Jose Barboza

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Instructor: Dr. Wei

Set: HW Set 2

Pack Pledge:

I have health given her received whoMuted and
on this assignment

Signature: JMB

Jose CHE 205 Problem Set 2 9/7/88

1) attached

2) 10.0% methyl alcohol, 75% mole methyl acetate
(mole)

15% mole acetic acid

used internet
to find
values
↓

100 mole sample → 10 mols methyl alcohol $\times 32.04 \text{ g/mol} = 320.4 \text{ g}$
75 mols methyl acetate $\times 74.08 \text{ g/mol} = 5556 \text{ g}$
15 mols acetic acid $\times 60.052 \text{ g/mol} = 900.78 \text{ g}$

mass fractions

(g/g)

methyl alcohol	$320.4 \div (320.4 + 5556 + 900.78) = 0.047$
methyl acetate	$5556 \div (320.4 + 5556 + 900.78) = 0.819$
acetic acid	$900.78 \div (320.4 + 5556 + 900.78) = 0.133$

Average molecular weight

$$(0.047 \times 32.04) + (0.819 \times 74.08) + (0.133 \times 60.052) = 70.2 \text{ g/mol}$$

Mass of sample with 25.0 kmol methyl acetate

25.0 kmol = 25,000 mols methyl acetate

methyl alcohol $25,000 / 75 = 333.33 \times 10 = 3333.3 \times 32.04 / 1000 = 106.79 \text{ kg}$
methyl acetate $25,000 / 75 = 333.33 \times 75 = 25,000 \times 74.08 / 1000 = 1852 \text{ kg}$
acetic acid $25,000 / 75 = 333.33 \times 15 = 5,000 \times 60.052 / 1000 = 300.26$

$106.79 + 1852 + 300.26 = 2259.05 \text{ kg}$

$\frac{3}{4} n_{\text{total}} = n_{\text{methyl acetate}} \quad n_{\text{total}} = \frac{25,000}{\frac{3}{4}} = \frac{100,000}{3} \text{ mols} \times \frac{70.2}{1000} = 2340 \text{ kg}$

Jose

3)

$$\frac{1 \text{ g-mol}}{6 \times 10^{23} \text{ molecules}}$$

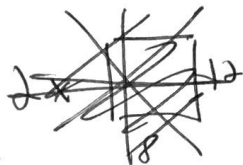
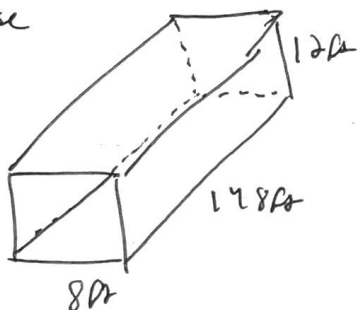
$$\frac{116}{453.593} =$$

$$\frac{116 \text{ mol}}{2.72 \times 10^{26} \text{ molecules}}$$

116 mol is 2.72×10^{26} molecules

4) Juse

a)



2 x 12 x 148



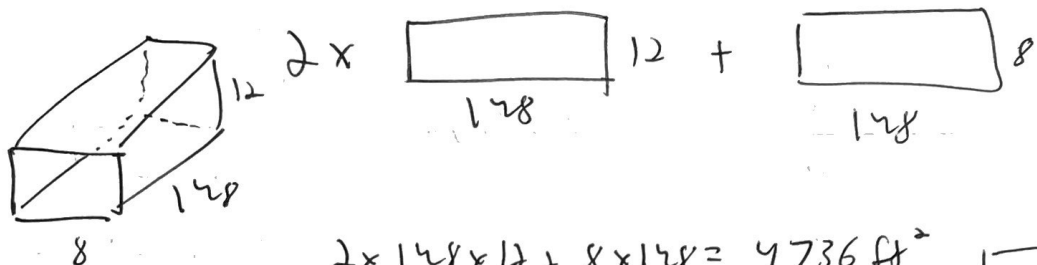
$$2 \cdot 8 \cdot 12 + 2 \cdot 12 \cdot 148 + 2 \cdot 8 \cdot 148 = \text{5920 ft}^2$$

$$\frac{5920 \text{ ft}^2}{25 \text{ ft}^2/\text{can}} = 236.8 \text{ cans} \rightarrow \boxed{237 \text{ cans}}$$

Dose

4)

a)



$$2 \times 128 \times 12 + 8 \times 128 = 4736 \text{ ft}^2$$
$$\frac{4736 \text{ ft}^2}{25 \text{ ft}^2/\text{can}} = 189.44 \text{ cans}$$

b)

Top 3 Hazards

1) Extremely Flammable aerosol

- in order to mitigate the chances of starting any aerosols, students can ensure no one is smoking, has any open flame, or is working with any electrical equipment that could give off a static discharge. This would help prevent ignition of aerosols.

2) Gas under pressure, may explode if heated

- to prevent explosion of cans, students can handle the cans safely and prevent cans from being heated through an external source (e.g. sun / fire).

3) Causes skin irritation

- in order to prevent skin irritation, students can wear long sleeve shirts, long pants, closed foot shoes, and gloves to avoid skin to paint/aerosol contact. Masks and eye protection should also be worn.

5) a) $SG = \frac{\rho}{\rho_{ref}}$ Just

$$1.24 = \frac{\rho}{62.43}$$

$$\rho = 77.4132 \text{ lb}_m/\text{ft}^3$$



or

$$m = SG \cdot \rho_{ref} \cdot V$$

$$V = 8 \times 8 \times \pi = 64\pi \text{ ft}^3$$

$$m = 1.24 \cdot 62.43 \times 64\pi = 15564.87 \text{ lb}_m$$

$$15564.87 \text{ lb}_m = 15564.8 \text{ lb}_f \rightarrow \boxed{15600 \text{ lb}_f}$$

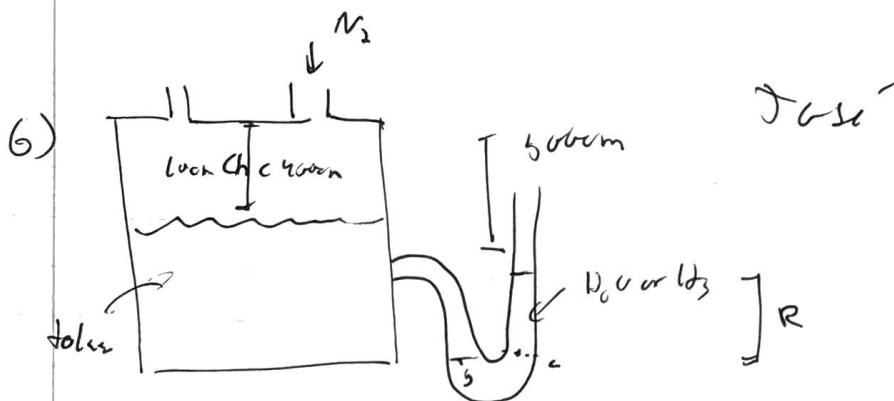
(3 sig figs)

b) pressure at bottom of tank

$$psig = \frac{SG \cdot \rho_{ref} \cdot g \cdot h}{g_c} = \frac{1.24 \times 62.43 \times 32.174 \times 8}{32.174}$$

$$= \boxed{619.31 \text{ psig}}$$

c) Because the density of molten chocolate is higher than that of H_2O , since an average human adult is 20% H_2O , reason should say that the worker should float. Despite this, it is entirely possible that 1) the worker inhaled a decent volume of chocolate after falling in, which would lead to respiratory problems and 2) due to the thickness of the liquid (molten chocolate) they would have been unable to swim, possibly resulting in injuries due to the equipment mixing the vat of chocolate.



$$h = 150 \text{ cm}$$

a) i)

$$500 - 150 = 350 \text{ cm at balance}$$

$$P_s = 350 \text{ cm} \times 9867 \text{ g/cm}^3 \times 9.8$$

$$c = \rho gh = 13575.898 \times 9.8 \times R$$

$$d = \rho gh = 867 \times 9.8 \times R \quad R = 150 \text{ cm}$$

$$R > 13575.898 \times 9.8 = 867 \times 9.8 \times 150$$

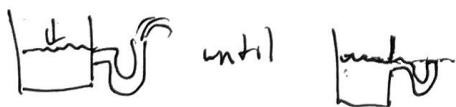
$$\boxed{R = 9.60 \text{ cm for } H_2}$$

$$R \times 997 \times 9.8 = 867 \times 9.8 \times 150$$

$$\boxed{R = 150 \text{ cm for } H_2O}$$

use H_2 as
you only have
a short wire,
instead of need
a meter long
tube

- b) If only using balance, the pressure would be equal on both sides of the barometer, so the liquid would flow out to equalize to the same height. you should use something heavier than balance, like H_2 , to keep it contained



- c) to keep from overflowing and keep gas so it doesn't combust with a gas

Honors Problem

Dosi

$$7.83 = \frac{7230 \text{ mm}}{7230 \text{ mm}}$$

$$\frac{7230}{7230} \times 4.8 \times e = 756$$

$$e = 0.010726$$

$$730 \times 28 \times 0.010726 = 24.33 \text{ mmHg}$$

$$756 - 24.33 = 731.67 \text{ mmHg, drop in pressure}$$

Jose Barbeito
9/7/22
Problem set 2, question 1

After taking my test, I learned much about how I prefer to learn in a classroom setting. The results that I received, a 3 for Active/Reflective, a 1 for Sensing/Intuitive, a 9 for Visual/Verbal, and a 9 for Sequential/Global, tell me the type of learner I am: an energetic, visual learner who would prefer to learn a process step by step. Looking into each of these four categories, I can further understand my preferences and change my actions within class to better suit my learning style.

My Active/Reflective score was a 3 (leaning active), which makes a lot of sense to me. I prefer being active in the way I think and interact with the class and others, but I also put a lot of effort into thinking deeply about certain topics on my own time. This balances out the two sides, but I lean more towards the active side because I'm a naturally energetic person, especially in the classroom setting. I'm excited to see what other classmates have an Active learning style, because I would prefer them in group works. I thoroughly enjoy working in a group where people are energetic, constantly bouncing ideas off one another.

My Sensing/Intuitive score was a 1 (leaning sensing). This one surprised me, but upon further reflection, makes a lot of sense. Being an engineer, I love order and straightforward processes. I much rather prefer a set way of doing things rather than being thrust upon a question and told to struggle with it until I figure it out. For scenarios like the latter, I end up remembering more the harsh struggle and my negative feeling about it than the actual process and the end result. This in turn means I end up not knowing how to apply the theory to another similar yet different problem. This observation would explain why I lean more Sensing than Intuitive. I believe that what balances out that side of me is a creative facet that I have cultivated throughout my life. I love creating things and am constantly searching for a creative outlet, whatever that may be. This, I believe, is the reason for my "intuitive" side, as most of the questions were geared towards your creative freedom within a scientific context, while I answered about my creative intuition within a life context.

The next category is no surprise. In the Visual/Verbal section, I am a 9 (leaning visual). I despise only listening to something being explained, and would much rather see it with my own eyes. Many times I have been able to copy someone's actions - whether it be a sport or a method of solving a problem - simply by observing them do it. I learn visually, and seeing charts and diagrams helps me to learn the subject matter tremendously.

The last section, Sequential/Global, gave me a score of 9 (leaning sequential). I had never heard of this category before, and learning that I was a sequential learner resonated true with me. Giving me a vague understanding of how the whole works does not help my brain to understand it. I am very easily able to match pieces together and see how they work in conjunction when I fully understand each detail. I refuse to leave any detail about the part unnoticed or misunderstood, and this usually leads me to either have an immense amount of knowledge about a very distinct part of the process or getting easily bogged down in the wrong details. It has helped me in the past and has hurt me as well.

Overall, this learning test was helpful. It gave me a new facet to see the way I learn and will surely help me understand how to be a better engineering student in the future at NC State.

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